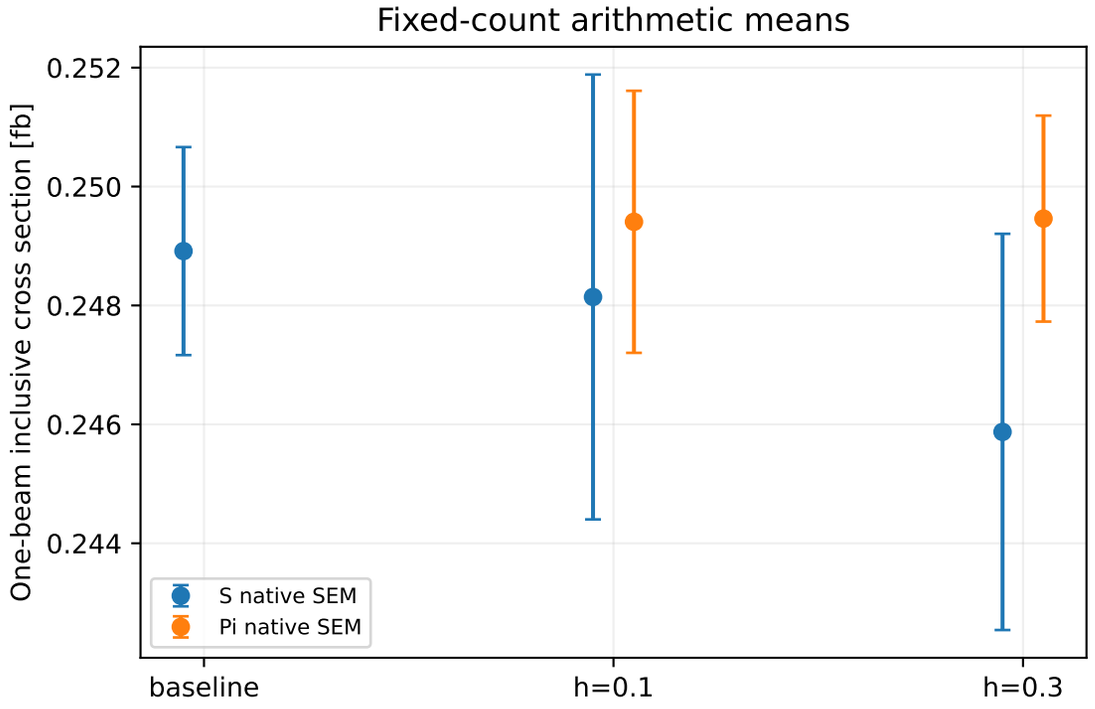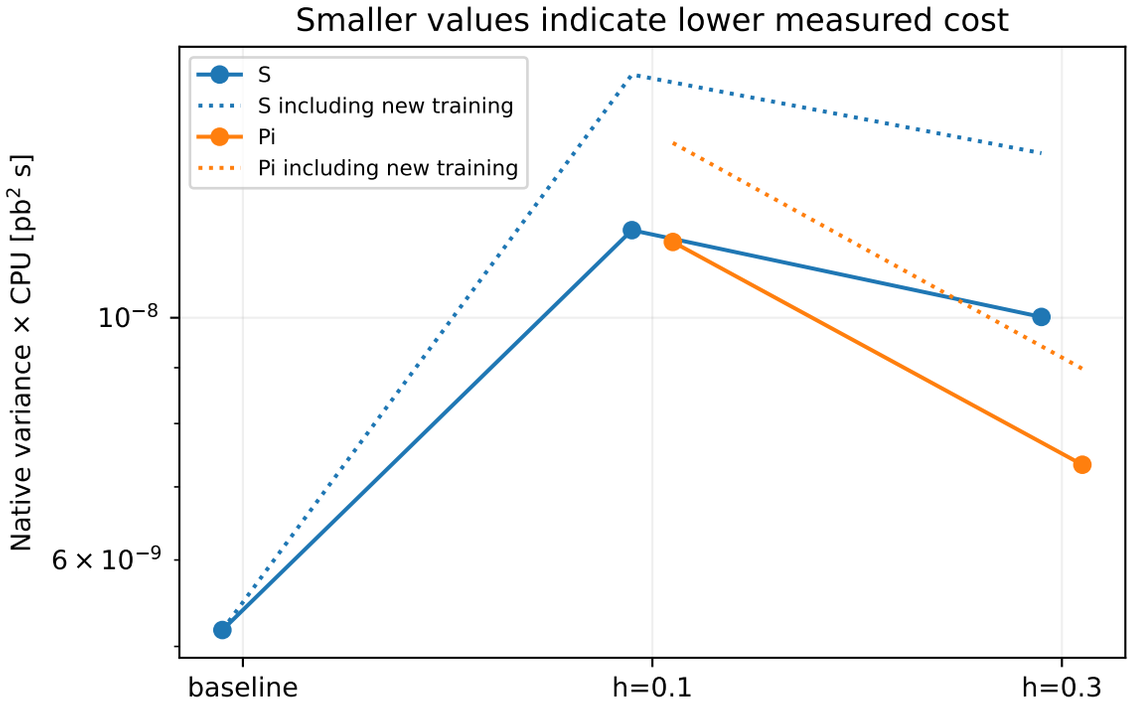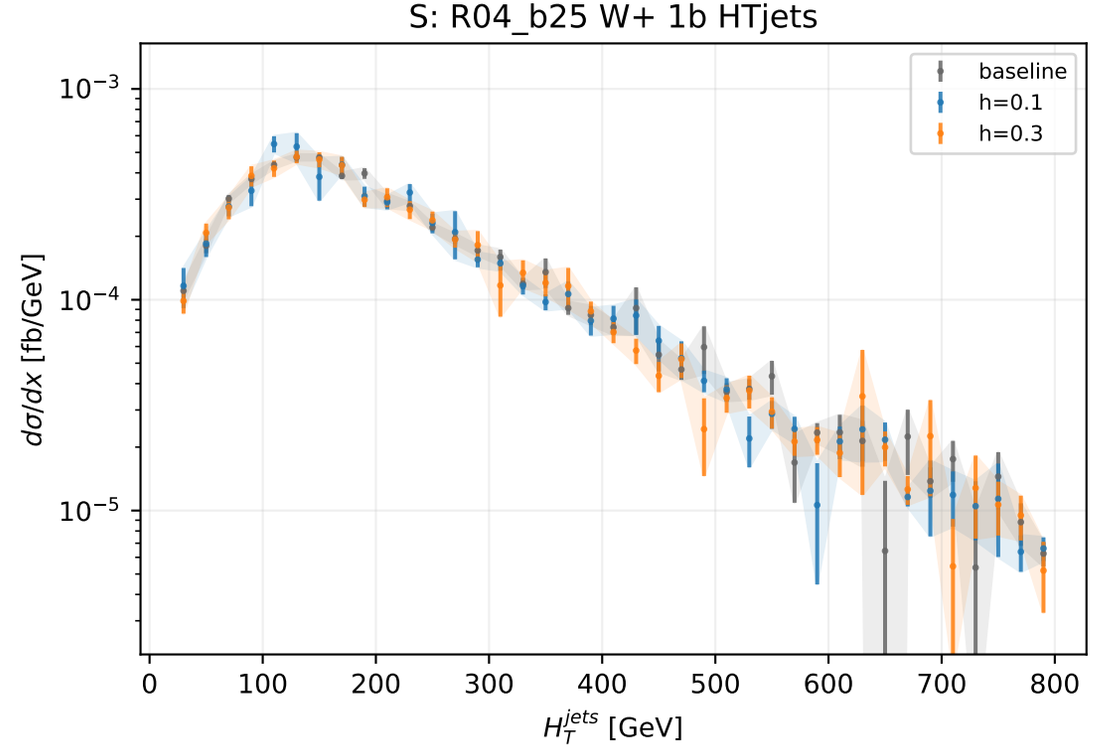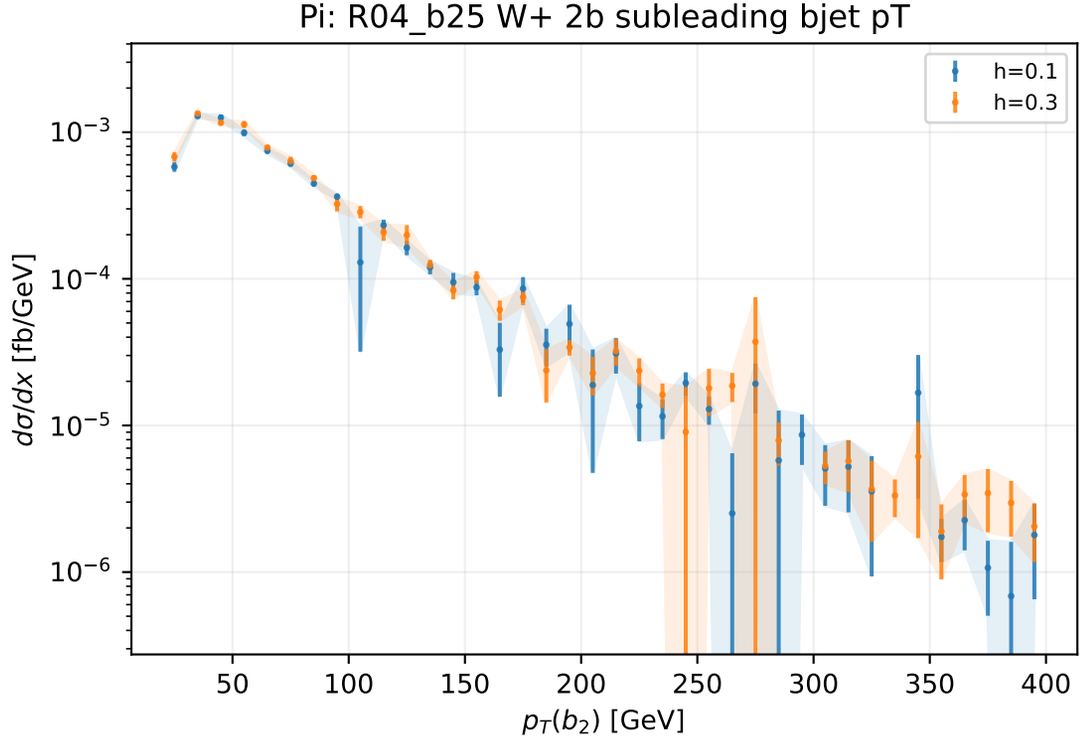Production hard-split diagnostic: central weights, one beam ordering
Legacy Pi reference excluded: production-real spin frames corrected